

LEUCHARS, United Kingdom

WMO: 031710

Lat: 56.377N

Lon: 2.863W

Elev: 12

StdP: 101.19

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -3.8	(c) -2.3	(d) -6.7	(e) 2.1	(f) -1.2	(g) -5.3	(h) 2.4	(i) 0.2	(j) 18.0	(k) 7.5	(l) 15.8	(m) 7.7	(n) 2.5	(o) 270	(p) 0.693

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 7.6	(c) 22.0	(d) 16.4	(e) 20.4	(f) 15.5	(g) 19.2	(h) 14.7	(i) 17.4	(j) 20.5	(k) 16.5	(l) 19.1	(m) 15.7	(n) 18.0	(o) 6.1	(p) 250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.2	11.5	18.7	15.4	10.9	17.7	14.6	10.4	16.8	49.1	20.4	46.3	19.2	44.1	18.0	21.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 13.9	(b) 12.1	(c) 10.6	(e) -6.6	(f) 25.2	(g) 2.4	(h) 2.0	(i) -8.3	(j) 26.6	(k) -9.8	(l) 27.7	(m) -11.1	(n) 28.8	(o) -12.9	(p) 30.3		
(a) 13.9	(b) 12.1	(c) 10.6	(e) -7.0	(f) 19.2	(g) 2.4	(h) 1.0	(i) -8.7	(j) 19.9	(k) -10.1	(l) 20.5	(m) -11.4	(n) 21.1	(o) -13.1	(p) 21.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.9	3.9	4.3	5.6	7.5	9.9	12.9	14.9	14.6	12.8	9.7	6.3	4.0
	DBStd	4.77	2.74	2.77	2.80	2.62	2.62	2.21	2.08	1.99	2.19	2.69	2.95	3.40
	HDD10.0	948	189	160	138	83	34	3	0	0	3	38	114	186
	HDD18.3	3446	447	392	394	324	262	164	109	116	166	266	361	443
	CDD10.0	547	0	1	2	8	31	88	151	144	87	30	3	1
	CDD18.3	4	0	0	0	0	0	0	2	1	1	0	0	0
	CDH23.3	18	0	0	0	0	1	2	10	6	1	0	0	0
	CDH26.7	1	0	0	0	0	0	0	1	1	0	0	0	0
Wind	WSAvg	4.8	5.4	5.5	5.5	4.8	4.6	4.4	4.1	4.1	4.5	4.9	4.8	5.1
Precipitation	PrecAvg	665	67	45	46	44	51	51	52	58	60	67	61	61
	PrecMax	991	167	142	102	102	152	107	146	144	145	170	194	155
	PrecMin	416	20	4	7	4	15	10	10	11	5	12	14	13
	PrecStd	116	29	31	23	25	28	25	28	34	34	39	35	30
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.3	13.0	15.2	17.6	20.2	22.8	24.6	23.6	21.9	17.7	14.2	13.0
		MCWB	10.3	10.7	10.7	12.2	14.5	16.1	18.1	17.4	16.7	14.1	12.2	11.5
	2%	DB	10.8	11.4	12.9	15.2	17.7	20.3	22.2	21.2	19.6	16.0	12.9	11.6
		MCWB	9.2	9.6	9.2	11.0	12.7	14.8	16.6	16.1	15.2	13.2	11.2	10.1
	5%	DB	9.5	10.2	11.6	13.6	16.0	18.6	20.5	19.8	17.9	14.9	11.8	10.4
		MCWB	7.9	8.3	8.5	9.6	11.6	13.8	15.7	15.3	14.0	12.6	10.1	9.1
	10%	DB	8.3	8.9	10.3	12.2	14.5	17.2	19.0	18.5	16.7	13.8	10.8	9.1
		MCWB	6.9	7.0	7.7	8.8	10.9	13.0	14.9	14.6	13.4	11.8	9.4	7.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.7	11.2	11.4	13.3	15.4	17.1	19.0	18.4	17.7	14.9	12.7	11.6
		MCDB	11.9	12.6	14.3	16.5	19.0	21.0	23.5	21.8	20.9	16.7	13.8	12.9
	2%	WB	9.4	9.7	10.0	11.5	13.5	15.6	17.6	17.2	16.0	13.9	11.5	10.3
		MCDB	10.6	11.2	12.4	14.4	16.6	18.9	20.8	19.8	18.3	15.4	12.6	11.5
	5%	WB	8.1	8.5	9.0	10.3	12.4	14.6	16.6	16.3	14.9	13.0	10.5	9.2
		MCDB	9.3	10.0	11.0	12.9	15.1	17.5	19.3	18.5	16.9	14.3	11.5	10.2
	10%	WB	7.0	7.2	7.9	9.2	11.4	13.7	15.7	15.6	14.1	12.2	9.6	8.0
		MCDB	8.2	8.6	10.0	11.6	13.7	16.3	18.1	17.5	15.9	13.5	10.7	9.0
Mean Daily Temperature Range	5% DB	MDBR	5.1	6.0	6.9	7.3	7.5	7.3	7.6	7.7	7.2	6.5	5.9	5.3
		MCDBR	6.2	6.7	8.4	9.4	9.7	9.8	10.0	9.3	9.0	7.2	6.4	6.3
		MCWBR	5.5	5.4	5.7	5.9	5.6	5.4	5.3	5.1	5.4	5.2	5.5	5.7
	5% WB	MCDBR	6.1	6.5	7.7	8.7	8.8	8.8	8.9	7.8	7.9	6.6	6.1	6.2
		MCWBR	5.6	5.5	5.7	5.7	5.6	5.4	5.3	5.0	5.4	5.2	5.5	5.7
Clear-Sky Solar Irradiance	taub	0.317	0.334	0.365	0.387	0.384	0.385	0.386	0.373	0.358	0.342	0.315	0.299	
	taud	2.272	2.366	2.313	2.282	2.339	2.364	2.368	2.414	2.465	2.443	2.338	2.234	
	Ebn at Noon	569	726	790	828	855	857	848	836	798	709	572	484	
	Edh at Noon	53	72	96	114	114	113	111	99	82	66	50	43	
All-Sky Solar Radiation	RadAvg	0.49	1.14	2.16	3.56	4.75	4.68	4.51	3.59	2.52	1.35	0.64	0.36	
	RadStd	0.07	0.10	0.27	0.38	0.30	0.39	0.39	0.25	0.16	0.16	0.07	0.04	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(47) Regional (44 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Nomenclature: See separate page